Portfolio Management

Report

FINA 6203: Investment Analysis – Fall 2025

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1- Investment objectives/benchmark and rationales

Capital growth:

Our objective is to invest in companies with strong earnings growth, durable business model and clear market leadership, focusing on long-term fundamentals rather than strict revenue-growth thresholds.

This approach targets stable, high-quality firms capable of delivering sustained compound growth. Companies such as Microsoft fir this profile, combining financial strength, competitive advantages and predictable earnings. Focusing on fundamentals allows flexibility across market conditions and supports steady long-term appreciation.

Diversification:

Our objective is to maintain exposure across multiple economic sectors to reduce concentration risk while keeping the portfolio aligned with long-term investment themes.

Sector diversification reduces portfolio volatility and aligns exposure with long-term structural themes. Technology provides innovation-driven returns, while healthcare and consumer staples add stability. ETFs broaden exposure and minimise idiosyncratic risk, lowering dependence on single-stock performance.

Benchmark:

Our objective is to evaluate the portfolio performance against S&P 500.

The S&P 500 is an appropriate benchmark because the portfolio primarily invests in large-cap US equities. Comparing performance to this index provides a realistic and consistent measure of whether out stock selection is outperforming or underperforming the broader US market.

Risk tolerance:

Our goal is to maintain a moderate level of risk by balancing high-growth, higher-volatility stocks with lower-beta defensive positions.

This mix aims to capture upside from growth sectors while mitigating downside exposure through stable, lower-volatility holdings. The resulting portfolio beta of about 0.35 confirms conservative market sensitivity and aligns with our team's moderate risk objective.

Rebalancing:

We want to follow a low-turnover approach unless risk exposure increases significantly, position sizes become overly concentrated, or market conditions justify an adjustment.

A low-turnover limits transaction costs and preserves a long-term investment horizon. Rebalancing is performed only when risk metrics deviate from target ranges, maintaining discipline and avoiding reactive trading.

Liquidity policy:

The objective is to maintain flexible liquidity based on market opportunities, holding enough cash to enter attractive positions when appropriate.

Dynamic liquidity management enables entry into attractive positions; however, excessive cash holdings can suppress returns. Balancing optionality with market exposure is essential for performance.

Asset allocation:

Our goal is to ensure no individual stock exceeds 10% of total portfolio.

This protects against concentration risk and ensures broad exposure to multiple sectors and companies, consistent with institutional portfolio management standards.

Review and adjustment:

Periodically review macroeconomic conditions, market indicators, quarterly earnings and risk levels. Adjust the portfolio only when justified by research or fundamental developments.

A structured review process ensures that portfolio positions remain consistent with underlying fundamentals and strategic objectives. It also mitigates the risk of biases or sentiment-driven decision-making by grounding adjustments in evidence-based analysis.

2- Investment strategies and/or investment policies statement

Our portfolio adheres to a core-satellite framework, with a clear and practical investment policy statement serving as the overarching strategy for allocating assets mainly to high-quality growth exposures. Broad-based index ETFs like VOO and IWM, which offer diversified exposure to U.S. large-cap stocks, international stocks, and U.S. small-cap stocks, respectively, make up the core allocation. By employing these broad index products, we lower the idiosyncratic risk associated with individual securities, maintain tracking error relative to the S&P 500 at a moderate level, and align the overall performance of the portfolio with the long-term trajectory of global equity markets.

On top of this core allocation, we build several satellite positions to reflect the team's constructive views on specific companies and sectors. We have established targeted positions in Apple (AAPL), Amazon (AMZN), iShares Russell 2000 ETF (IMW), Microsoft (MSFT), Eli Lilly (LLY), Coca-Cola (KO), Lockheed Martin (LMT) and Vanguard S&P 500 ETF (VOO). Through these individual securities and sector ETFs, we intentionally increase exposure to themes such as artificial intelligence, cloud computing, healthcare innovation, consumer brands, and defense. While aiming to produce better long-term, risk-adjusted returns than our benchmark, this design enables us to maintain wide diversification and strong risk control.

When it comes to investment strategy, we typically prefer allocating to reputable businesses that have strong free cash flow, stable balance sheets, and long-term competitive advantages. We have included a few defensive and value-oriented assets in addition to the portfolio's comparatively high percentage of large-cap technology and pharmaceutical stocks, which show a certain growth-oriented bias. These comprise stocks in industries like defense and consumer staples. This strategy seeks to capitalize on market surges while acting as a safety net during corrections or periods of increased volatility. Conversely, we actively avoid companies with weak fundamentals, unclear profit models, or highly speculative characteristics. Instead, we favor long-term holdings in firms demonstrating stable

earnings histories, disciplined capital allocation, and relatively clear drivers for future profit growth. This strategy enhances the overall quality and sustainability of the portfolio.

3- The investment decision process for identifying, selecting and monitoring securities

The investment decision process involves a multi-layered approach for identifying, selecting, and monitoring securities. Identification and selection are driven by both long-term strategic goals and short-term tactical opportunities. A core foundation is established through broad market diversification, which is why we invested the majority of our deployed capital in VOO; it provides stable, low-risk exposure to the U.S. market and serves as the base for the rest of our strategy.

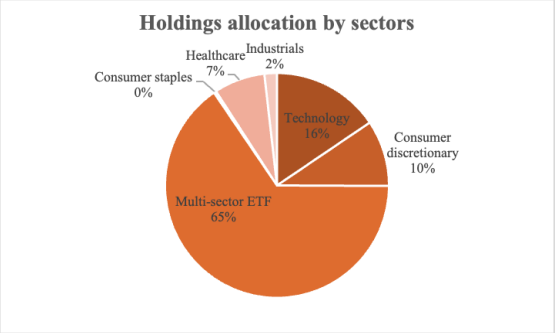
A key element of our approach was the deliberate decision to invest only 55% of the portfolio and keep 45% in cash. This was intentional because market conditions at the start of the simulation showed elevated valuations and short-term volatility. Maintaining a large cash position allowed us to limit drawdowns, avoid buying at extended prices, and preserve flexibility for future opportunities. Although not standard among groups, this approach best aligned with our risk-management priorities.

Satellite allocations were selected for high-conviction themes, targeting secular growth leadership, diversified tech exposure, and defensive healthcare stability. Monitoring and risk management remained continuous, using sales to reduce over-concentration, eliminate functional overlap, and capture gains with plans to repurchase at lower targets. The process further integrates structured short-term tactical trades with defined theses, clear entry points, stop-loss parameters, and profit-taking plans, ensuring rapid adjustments to lock in strength and reduce immediate risk.

4- Portfolio holdings, value and performance history

A- Portfolio holdings

Symbol	Description	Market Value	Weight	Sector
AAPL	Apple Inc.	$138,775.00	2.96%	Technology
AMZ	Amazon.com, Inc	$457,174.20	9.52%	Consumer discretionary
IWM	iShares Russell 2000 ETF	$247,300.00	5.10%	Multi-sector ETF
KO	The Coca-Cola Company	$14,576.00	0.34%	Consumer staples
LLY	Eli Lilly and Company	$383,205.98	7.37%	Health care
LMT	Lockheed Martin Corporation	$90,832.00	1.81%	Industrials
MSFT	Microsoft Corporation	$619,012.50	12.59%	Technology
VOO	Vanguard S&P 500 ETF	$2,937,265.00	60.32%	Multi-sector ETF

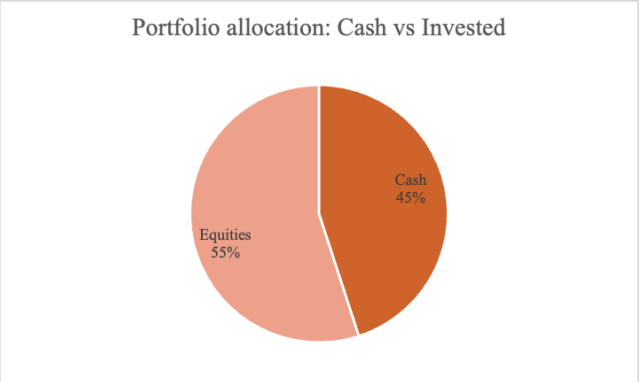


The portfolio is dominated by VOO (60.32%), which provides broad U.S. market exposure and reduces idiosyncratic risk. The large allocation aligns with our core–satellite structure, where VOO serves as the foundation of the portfolio. Microsoft (12.59%) and Amazon (9.52%) offer growth potential through large-cap technology exposure, while Eli Lilly (7.37%) provides stability from healthcare sector. Smaller positions in AAPL, LMT, IWM, and KO add diversification across sectors. Overall, the allocation reflects a moderately diversified portfolio with a tilt toward large-cap U.S. equities.

B- Portfolio value

Total Market Value	$4,888,140.68
Total Cost Basis	$4,758,037.81
Total Unrealized P/L	$130,102.87
Return on invested capital	**2.73%**

Portfolio Value	$9,931,700.15
Beginning Value	$10,000,000.00
Total Return	**-0.68%**



The portfolio allocated 55% to equities and 45% to cash. While this aligned with our liquidity objective, the high cash balance became the primary source of underperformance by materially reducing participation in market gains. Equity results were modest—some securities posted small gains (AAPL, LLY, KO, VOO) while others showed slight declines (MSFT, AMZN, IWM, LMT), indicating that stock selection contributed only marginally to overall performance.

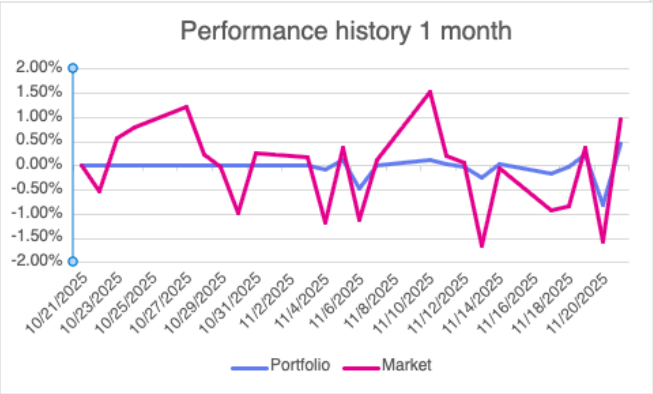
Performance attribution confirms that the allocation effect dominated. Holding 45% in cash created significant cash drag, limiting exposure to market appreciation. The selection effect was relatively positive, as the invested portion returned 2.73% *(Exhibit 1)*. However, the interaction effect was negative: limited and low-beta equity exposure reduced participation in market upswings relative to the S&P 500.

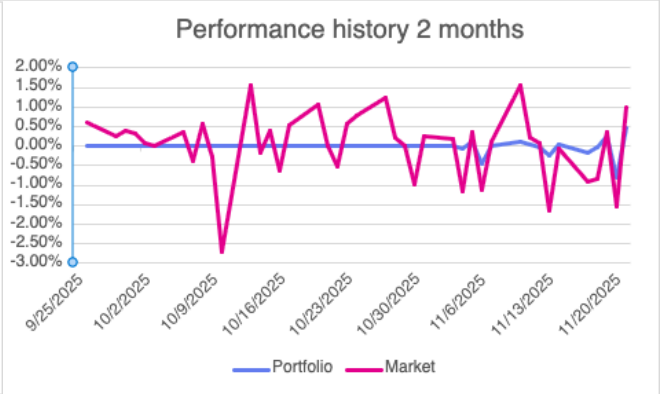
Consequently, the total portfolio value fell to $9,931,700.15 on 11/21/2025, resulting in a –0.68% overall return despite gains within the invested portion. This shortfall reflects the opportunity cost of excess liquidity rather than deficiencies in security selection. A lower cash allocation would have allowed stronger alignment with market performance.

C- Performance history

	Portfolio	**Market**
Total return	-0.683%	-0.026%
Mean daily return	-0.017%	0.003%

The table shows that the portfolio underperformed the market over the evaluation period, generating a total return of –0.683% compared with the market's -0.026%. This underperformance is also reflected in the mean daily return, where the portfolio delivered –0.017% per day versus the market's 0.003%. The consistently negative daily mean indicates a gradual and persistent pattern of mild underperformance. Both cumulative and average daily returns therefore show that the portfolio lagged the benchmark throughout the evaluation window, largely due to limited market exposure and a low effective beta that restricted participation in market gains.





Over the past two months, the portfolio displayed very low volatility, with daily returns remaining close to zero while the market experienced sharp swings. As a result, the portfolio captured only a

limited portion of the market's upward movements. This pattern persisted in the most recent month: although the portfolio moved in the same general direction as the benchmark, its fluctuations were far smaller. This muted sensitivity reflects a low effective beta, explaining why the portfolio underperformed during market upswings while offering only modest downside protection during periods of decline.

5- Portfolio characteristics

A- Risk measures

	Portfolio	**Market**
Variance	3.099E-06	7.515E-05
Standard Deviation	0.002	0.009
Beta	0.108	1.00

The portfolio produces two different beta values depending on the calculation method.

The first beta, $\beta \approx 0.108$ *(Exhibit 2)*, comes from a regression of the portfolio's daily returns against the market. This estimate is heavily reduced by the fact that 45% of the portfolio is allocated to cash, which has a beta of zero. Combined with the short evaluation window and the limited variability in daily return, the regression beta therefore understates the portfolio's true market sensitivity.

The second beta, $\beta = 0.512$ *(Exhibit 3)*, is calculated as the weighted average of the individual stock betas. This metric reflects the systematic risk of the invested portion of the portfolio and aligns with the beta reported by Stock Trak. Because it excludes cash holdings, it reflects the true systematic risk of the invested equity component.

B- Performance measures

	Portfolio	**Market**
Jensen Alpha	-3.16E-04	2.92E-04
Sharpe Ratio	-3.994	-0.053
Treynor Ratio	-0.065	-0.0005

The performance metrics clearly show that the portfolio underperformed the market across all risk-adjusted dimensions. The portfolio's Jensen's Alpha (–3.16E-04) indicates that, after controlling for systematic risk, the portfolio earned less than what CAPM would predict. In contrast, the market's positive alpha (2.92E-04) suggests a slight outperformance relative to expectations.

Risk-adjusted ratios further reinforce this underperformance. The portfolio's Sharpe Ratio (–3.994) is significantly below the market's –0.053, meaning that once returns are adjusted for volatility, the portfolio delivered much weaker performance relative to the risk-free rate. This reflects the substantial drag created by the low equity exposure and large cash position.

Similarly, the Treynor Ratio (–0.065), compared with the market's –0.0005, shows that the portfolio did not generate adequate returns for the small amount of systematic risk it carried. With a low beta, the portfolio captured limited upside yet still delivered negative excess returns.

Overall, all three measures point to the same conclusion: the portfolio was far less efficient than the benchmark, failing to compensate investors for either total risk or systematic risk. These results are consistent with a portfolio that remained under-invested and overly conservative throughout the evaluation period.

C- Evaluation of objective achievements and key reasons

Objectives	Objective achievements	Reasons
Capital growth	Not achieved	High-quality stock selection generated gains, but the 45% cash allocation significantly limited participation in market appreciation, creating a material opportunity cost relative to the benchmark.
Diversification	Achieved	The portfolio maintained balanced exposure across sectors and utilised ETFs to reduce idiosyncratic and concentration risk. Despite a large core allocation to VOO, the portfolio demonstrated effective diversification across sectors and instrument.
Benchmark comparison	Not achieved	The portfolio materially underperformed the benchmark (S&P 500), primarily because the portfolio carried substantially less market risk than the benchmark due to high cash holdings and a low effective beta.

Risk tolerance	Achieved	The portfolio's low beta and allocation to defensive sectors aligned with the stated moderated-risk objective.
Rebalancing	Achieved	Turnover remained low and consistent with a long-term, fundamentals-driven investment approach.
Liquidity policy	Not achieved	Excessive cash holdings were maintained throughout the period, restricting market participation and resulting in liquidity levels higher than intended.
Asset allocation limit	Achieved	All positions remained below 10% threshold, preventing excessing concentration in a single security.
Review and adjustment	Achieved	Portfolio decisions adhered to fundamental analysis, with no evidence of reactive or sentiment-driven trading.

6- Team organization and operations

The team collaborated through clearly defined roles that supported both strategy and execution. Lisa Bencivenga, as Portfolio Analyst and Strategy Advisor, researched the optimal number of shares for each security, contributed to shaping the overall portfolio strategy, and provided objective, data-driven insights. Youmei Bian, serving as Trader and Financial Analyst, executed trades, facilitated portfolio rebalancing, and identified appropriate entry and exit points. Winston Essibu, acting as Portfolio Strategist and Quality Reviewer, led discussions on portfolio structure, researched potential securities to strengthen the portfolio, and verified all calculations for accuracy and consistency. As Team Lead and Communications Coordinator, Anagha Athreyas liaised with the professor, developed detailed deliverable outlines, and organised team activities. Finally, Jaanvi Bajaj contributed to the investment process by assisting in the identification, selection, and monitoring of securities and supporting the documentation of the team's portfolio decisions.

Suggestions for future classes

Suggestions for future classes include maintaining the emphasis on using an investment policy statement to ensure a coherent strategic approach, holding weekly meetings to stay aligned and up to date, and continuing active research on economic events that may affect portfolio performance. However, several improvements could strengthen the learning experience. Providing clearer expectations regarding cash allocation would support better stock selection and overall portfolio

optimisation. Adopting a more time-sensitive workflow would also help students operate more effectively within a short evaluation period. Finally, incorporating additional analytical tools could enhance both decision-making and the depth of portfolio analysis.

7- Conclusion

Our portfolio management approach remained broadly consistent with the investment policy statement. We maintained sector diversification, adhered to a disciplined selection and monitoring framework, and focused on long-term fundamentals when adjusting positions. This structured process ensured coherence between our stated objectives and the implementation of our strategy.

However, despite strong procedural alignment, the portfolio underperformed the benchmark. The primary driver of this underperformance was the high cash allocation, which materially limited market participation and diluted gains from well-chosen equity positions. The portfolio's low effective beta prevented it from capturing market upswings, while the short evaluation window offered limited opportunities to gradually deploy cash or adjust exposures as conditions evolved.

Looking ahead, stronger performance would require more efficient capital deployment and greater emphasis on capturing short-term market opportunities within the broader long-term framework. Reducing excess liquidity, selectively increasing exposure to higher-beta assets, and integrating more dynamic allocation tools could help future teams enhance benchmark-relative returns.

Overall, the team successfully applied a coherent strategy, while also identifying clear avenues for refinement in future iterations of this project.

8- Exhibits

Exhibit 1: Returned on invested capital

Symbol	Description	Total price	Market Value
AAPL	Apple Inc.	$135,210.00	$138,775.00
AMZN	Amazon.com, Inc	$442,650.60	$457,174.20
IWM	iShares Russell 2000 ETF	$240,500.00	$247,300.00
KO	The Coca-Cola Company	$14,004.00	$14,576.00
LLY	Eli Lilly and Company	$358,253.21	$383,205.98
LMT	Lockheed Martin Corporation	$96,700.00	$90,832.00
MSFT	Microsoft Corporation	$616,692.00	$619,012.50
VOO	Vanguard S&P 500 ETF	$2,854,028.00	$2,937,265.00
Portfolio		$4,758,037.81	$4,888,140.68

Exhibit 2: Beta using a regression of the portfolio's daily returns

Total return	-0.683%
Mean daily return	-0.017%
Standard Deviation	0.002
Covariance	8.101E-06
Variance	3.099E-05
Beta	0.108

Exhibit 3: Weighted-Average beta of the individual stocks

Weighted-Average Beta		
	Beta	**Weight**

AAPL	0.608	2.96%
AMZN	1.552	9.52%
IWM	1.397	5.10%
KO	0.404	0.34%
LLY	0.962	7.37%
LMT	-0.516	1.81%
MSFT	0.296	12.59%
VOO	0.985	60.32%
Stocks Beta	0.931	55%
Portfolio Beta	**0.512**	

The 55% figure reflects the proportion of the portfolio invested in equities. In calculating the portfolio beta, this invested share is applied to the weighted-average beta of the securities, whereas the 45% held as cash contributes a beta of zero.